

# Edward Denton



(Not on public CV)



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24edden

## Education

### University of Warwick

2023-2026

**Computer Science BSc** | *Warwick Award (Gold)*

- Current Grade: **First**
- Relevant Modules: Functional Programming (**Haskell**), Data Structures (**Java**), Computer Organisation and Architecture (**C**), Algorithmic Graph Theory, Complexity of Algorithms, Neural Networks, Databases

### King's School Grantham

2016-2023

- **A-A\*** in A-Levels (Maths, Further Maths, Physics, and Computer Science)
- **Computer Science Club**: Founded a club to support the extra-curricular development of students

## Experience

### Arm

June 2025 - September 2025

*SWE Intern (Fast Models) | Return Offer*

- Implemented features of a **CPU component generation script** for CPU simulation, including generating test cases
- Traced Windows test skips to unguarded Linux commands in the **Makefile**, adding platform branching
- Automated test migration from proprietary framework to **Robot Framework**, improving developer experience

### University of Warwick Computing Society

March 2024 - Present

*Academic Officer*

- Used pedagogical principles in producing resources for a 4-session **C Course** aimed at first year CS students
- Founded and led the **XSoc LeetCode Workshops**, a 6-week collaboration with CodeSoc, for technical interview prep
- Produced and presented **revision lectures** (e.g. for algorithms and automata) for exams and class tests

### Mystifine Productions (Game Development)

June 2020 - November 2020

*Programmer and Administrator*

- Built a Lua database on Roblox's DataStore API for player and guild data at **18k peak concurrent players**
- Applied best practices (DRY, YAGNI, Design Patterns) to enhance code readability and adaptability
- Managed a community of over **30k members**, addressing player feedback for continuous improvement

## Projects

### WildFire Modeller and Predictor (ICHack Win) | *Pixi, Flask, React, Matplotlib, Python, js, HTML, CSS*

- Worked in a 6-person team to produce a stochastic **percolation-based simulation** of wildfires in California
- Implemented Monte-Carlo simulations to produce a probability map for evaluating fire mitigation strategies
- Additionally applied TFT models to forecast wildfire spread overtime, using real-world datasets
- Produced in **24 hours for ICHack'26** and **won HRT's prize** for "Best use of Data for Predictions & Decision Making"

### CSNotes Project (Project Lead) | *Flask, Shadcn (and React), UV, Docker, KaTeX, Gemini (LLM)*

- Integrated LLMs (Gemini) to generate AI summaries of module reviews, with **caching** to avoid redundant API calls
- Designed the UI using **Shadcn** and **Tailwind**, including display themes (e.g. high-contrast) saved in local storage
- Built **interactive quizzes** with matching, multi-answer, and drag-to-sort questions, plus markdown/LaTeX rendering

### Chess and AI | *Python, Pygame, timeit*

- Built a chess AI with **minimax and alpha-beta pruning**, benchmarked with timeit to cut move-search time
- Designed a custom **event-driven architecture** in Pygame to support the game UI, menu, and forms system
- Implemented full chess rules including castling, en passant, promotion, and checkmate detection

### Logo Parser and Simulator | *Haskell, Gloss*

- Parsed Logo source into an intermediate representation with **Megaparsec**, handling case-insensitive keywords
- Interpreted the IR into a state ADT via the **State monad**, animating turtle movement with Gloss
- Built a CLI entry point surfacing positioned parse errors and visualising simulations from .hogo files

**Other projects** available at <https://github.com/24edden>

## Technical Skills

**Languages**: Lua, C, Haskell, Rust, Python, C++, Java, JavaScript, HTML, CSS

**Tooling**: Git, GitHub, Gerrit, Docker, Portainer, Figma, Jira

**Frameworks/Tech**: Flask, jQuery, Gloss, node.js, PyGame, Raylib

## Interests

Climbing (Bouldering), Calisthenics, EdTech/Pedagogy, Game Development, Competitive Programming, Research